A +- Understanding and Maintenance of Networks

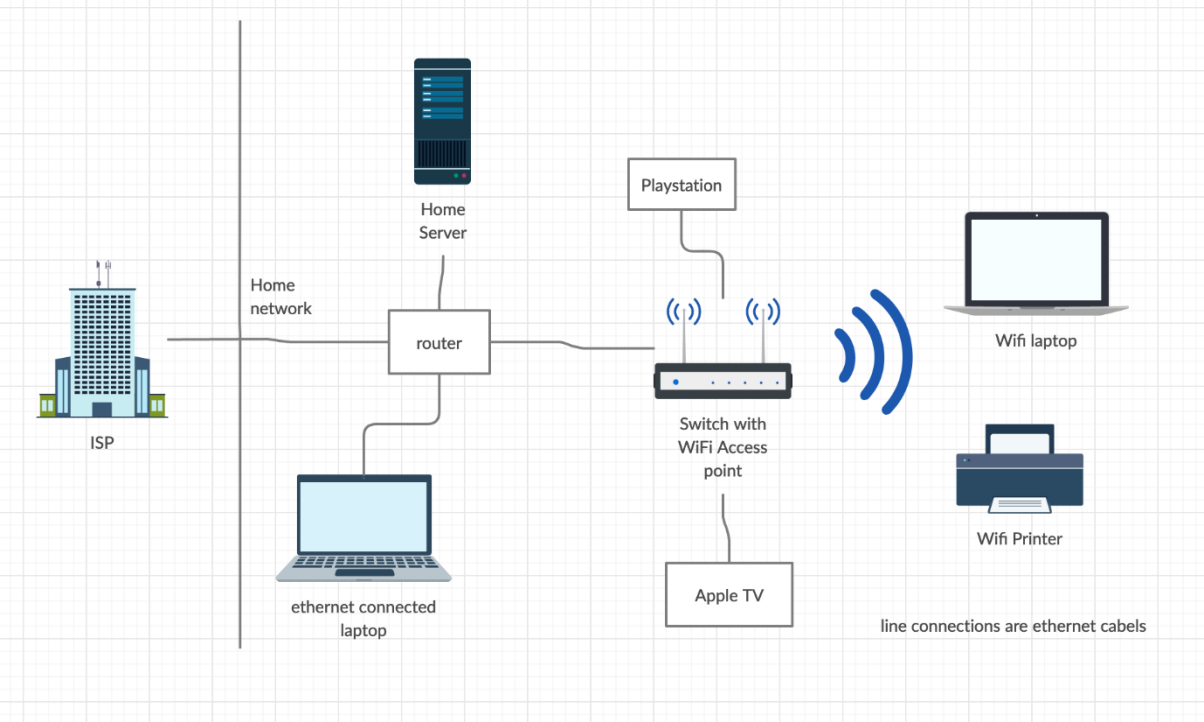
Day – 1

Introduction to Networking Hardware Theory : Overview of modems , routers ,switches ,access points and network cables

TASKS – 1

Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

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The internet connection comes from the Internet Service Provider (ISP) to the modem. The modem connects to the router, which directs network traffic. The switch provides additional wired connections, and the access point provides Wi-Fi coverage for devices such as laptops and smartphones.

TASK -2

List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

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1. Modem

Function: Connects your home network to the internet service provided by your ISP.

Practical example: A GTPL internet connection uses a compatible modem or modem-router device to provide internet access.

2. Router

Function: Connects different networks and directs data packets to their destinations. A home router usually also provides Wi-Fi and shares the internet connection.

Practical example: Your smartphone and laptop use your home router to access the internet.

3. Network switch

Function: Connects multiple wired devices in the same local area network (LAN) and forwards Ethernet frames to the appropriate ports.

Practical example: An office uses a switch to connect several desktop computers and printers.

4. Access point (AP)

Function: Allows wireless devices to connect to a wired network using Wi-Fi.

Practical example: A school installs access points in classrooms to provide Wi-Fi to students.

5. Network cable

Function: Carries data between wired networking devices.

Practical example: An Ethernet cable connects a computer to a switch or a router to a switch.

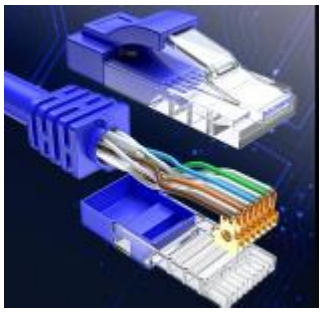
TASK - 3

Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

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1. ETHERNET Cable (UTP)

   Used to connect computers, routers, and switches in a wired LAN. Cat5e and Cat6 are common examples.



2. Coaxial cable

   Used for cable television and some cable-internet connections.



3. Fiber-optic cable

   Uses light to transmit data at high speeds over long distances, commonly for broadband internet and telecommunications.



TASK - 4

Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

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Feature	Router	Switch
Main Purpose	Connects Different Networks	Connect devices with a LAN
Data Handling	Uses IP to route Packets	A layer 2 switch uses a MAC address to transfer FRAMES
Typical Location	Between HOME, Office	HOME, OFFICE and Computer lab LAN
Example	Home wifi Router	8 port Ethernet Switch